

AAKASH SIRICILLA

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EDUCATION

New Jersey Institute Of Technology , Master of Science in Computer Science	Sep 2023 – May 2025
Coursework: Distributed Systems, Deep Learning, Machine Learning & MLOps	GPA: 3.93/4.0
Kommuri Pratap Reddy Institute Of Technology , B.Tech. Computer Science & Engineering	Aug 2019 – May 2023
Coursework: Cloud Computing, Data Structures & Algorithms, Operating Systems, Database Systems	GPA: 8.55/10

TECHNICAL SKILLS

Languages	Java, Python, TypeScript, SQL (PostgreSQL, MySQL), C/C++
Frontend	React, Next.js, Tailwind CSS, HTML, CSS
Backend & APIs	Spring Boot, FastAPI, REST APIs, WebSockets, JWT Authentication, Async Processing
Databases & Messaging	PostgreSQL, MySQL, Redis, Qdrant, FAISS
Cloud & DevOps	Docker, Kubernetes, CI/CD, AWS, Git, GitHub Actions
Systems & Tools	Celery, MLflow, Prometheus, Grafana, Pandas, NumPy

WORK EXPERIENCE

Spectrum Capital NJ	May 2025 – Present
Software Engineer, Distributed Systems	<i>New Jersey, USA</i>

- Engineered centralized financial operations dashboards using Java, Spring Boot, React, and PostgreSQL, consolidating portfolio tracking and client workflows to reduce manual reconciliation effort by 40%.
- Developed high-throughput REST APIs and optimized complex PostgreSQL query pipelines, cutting average response latency by 35% and enhancing throughput for data-intensive internal services.
- Automated end-to-end financial data ingestion and validation workflows via Python and SQL ETL pipelines, accelerating report generation by 60% while ensuring downstream data integrity.
- Architected role-based workflow orchestration and audit-tracking modules, implementing granular access controls and transactional logging that slashed issue investigation time by 45%.
- Designed and deployed containerized Spring Boot backend services on AWS using Docker and PostgreSQL, standardizing deployment environments, improving operational scalability, and reducing release inconsistencies across internal engineering systems.

New Jersey Institute of Technology	Mar 2024 – May 2025
Research Assistant	<i>Newark, NJ, USA</i>

- Optimized PostgreSQL and MySQL queries across 120K+ records by redesigning joins, refining retrieval logic, and improving indexing strategies, reducing query execution time by 40% and improving backend responsiveness for internal research applications.
- Automated Python-based ETL and preprocessing pipelines for recurring datasets, reducing manual processing effort by 60% and improving the consistency and reliability of downstream reporting workflows.
- Built internal data pipelines and reporting dashboards to track operational and model performance metrics, shortening reporting turnaround time by 30% and improving visibility into system performance for research stakeholders.

Cognier Insights	Apr 2023 – Sep 2023
Software Engineer	<i>Hyderabad, India</i>

- Developed customer-facing CRM modules and workflow automation features using Java, Spring Boot, React, and PostgreSQL, enabling streamlined lead management, client onboarding, and operational task tracking that reduced manual coordination effort by 35%.
- Designed and developed REST APIs, service-layer validation, and PostgreSQL-backed data access logic for data-intensive workflows, improving retrieval performance by 25% and increasing reliability of internal transaction flows.
- Created reusable React UI components and integrated them with Spring Boot services, reducing feature delivery time by 30% and improving consistency and maintainability across shared frontend modules.

PROJECTS

SyncStream - Real-Time Collaborative Streaming Platform | Java, Spring Boot, Next.js, WebSockets, PostgreSQL, Docker | [GitHub](#)

- Built a full-stack real-time collaboration platform with invite-based session flows, persistent room state, and low-latency synchronization between users using Spring Boot, WebSockets, and Next.js.
- Dockerized frontend and backend services, structured the project for local and cloud deployment, and implemented scalable client-server communication patterns for live session updates.

DocuMindAI - Enterprise Document Intelligence Platform | FastAPI, React, Qdrant, Redis, Docker, MLflow | [GitHub](#)

- Developed a scalable full-stack document platform with asynchronous ingestion, semantic search, and contextual chat, reducing document lookup time by nearly 70%.

- Implemented hybrid retrieval, background task execution with Celery and Redis, and modular service orchestration to support multi-document workloads and extensible backend architecture.

SentinelAI - Real-Time Network Monitoring Platform | Python, FastAPI, Redis, TimescaleDB, Next.js, Docker
| [GitHub](#)

- Engineered a real-time event processing pipeline with streaming analytics, Redis-backed state management, and sub-500ms request handling for high-volume anomaly detection workflows.
- Built a responsive monitoring dashboard with WebSocket-powered updates, replay controls, and live analytics views, improving visibility and investigation speed for operational events.